


Figure 1.36

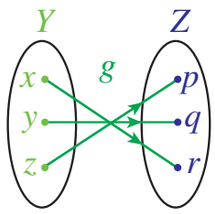



Figure 1.37

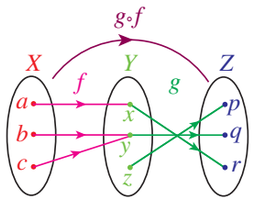



Figure 1.38

Now we can define a function $h : X \to Z$ in a natural way. To find the image of $a$ under $h$, we first see the image of $a$ under $f$; it is $x$; then we see the image of this $x$ under $g$; this is $r$. That is, $h(a) = r$. Similarly, we declare $h(b) = q$ and $h(c) = q$. In this way we can define a new function $h$. This $h$ is called the composition of $f$ with $g$.

**Definition 1.8**

Let $f : X \to Y$ and $g : Y \to Z$ be two functions. Then the function $h : X \to Z$ defined as $h(x) = g(f(x))$ for every $x \in X$ is called the ***composition*** of $f$ with $g$. It is denoted by $g \circ f$ (Read this as $f$ composite with $g$). (See Figures 1.38 and 1.39.)

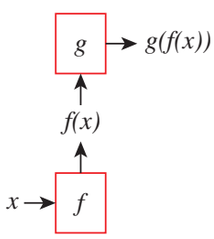



Figure 1.39

We can note that the range of $f$ need not be $Y$. If $f : X \to Y_1, g : Y_2 \to Z$ and $Y_1 \subseteq Y_2$, then also we can define $g \circ f$; we can take $Y_2$ as the co-domain of $f$ and use the same definition. So we can define $g \circ f$ if and only if the range of $f$ is contained in the domain of $g$.

**Example 1.25** Let $f = \{(1, 2), (3, 4), (2, 2)\}$ and $g = \{(2, 1), (3, 1), (4, 2)\}$. Find $g \circ f$ and $f \circ g$.

**Solution:**

To check whether compositions can be defined, let us find the domain and range of these functions.

Domain of $f = \{1, 2, 3\}$, Range of $f = \{2, 4\}$, Domain of $g = \{2, 3, 4\}$ and Range of $g = \{1, 2\}$. Since the range of $f$ is contained in the domain of $g$ we can define $g \circ f$; so as to find the image of 1 under $g \circ f$, we first find the image of 1 under $f$ and then its image under $g$. The image of 1 under $f$ is 2 and its image under $g$ is 1. So $(g \circ f)(1) = g(f(1)) = g(2) = 1$.